Mohammed Azhar Khan

AI / Machine Learning Engineer

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Professional Summary

AI/ML Engineer (B.E. in AI & ML, 2026) who reviews and corrects LLM-generated code at Invisible Technologies for benchmarking programs run by frontier AI labs. Builds and deploys complete ML systems, including a LightGBM demand forecaster with drift monitoring and automatic retraining, and a multi-agent research assistant on LangGraph. Works mainly in Python with scikit-learn, PyTorch, FastAPI, PostgreSQL and Docker.

Experience

Data Annotator / AI Trainer Oct 2025 – Present
Invisible Technologies *Remote*

- Review and grade AI-generated Python code for LLM benchmarking projects run for a frontier AI lab, and rewrite the reference solution whenever the model's logic or output is wrong.
- Analyse live conversations from a production virtual agent, group the failures by cause and feed them back as retraining data, which lowered the agent's NLU fallback rate.

Python with Generative AI Intern Jan 2026 – May 2026
Dhee Coding Lab *Onsite*

- Learned Python and generative AI (prompt engineering, calling LLM APIs) in hands-on lab sessions.
- Guided students through small projects built on AI APIs, from the first spec to a working demo.

Projects

NEM Forecast: Electricity Demand Forecasting System Live Demo | GitHub
Python, LightGBM, FastAPI, PostgreSQL, Docker, Evidently, GitHub Actions, Next.js

- Forecasts Victoria's half-hourly electricity demand from 30 minutes to 24 hours ahead using 3 years of public AEMO market data. Cut mean absolute error by 30% against the best of four baselines (351 MW vs 501 MW) over a full year of walk-forward backtests, and beat every baseline at all 48 horizons.
- Serves forecasts from a FastAPI service in Docker, released through a pipeline of 365 tests, strict type checks and security scans. Live forecasts are scored against actual demand (error within 3% of the backtest), drift is tracked with Evidently, and a retrained model goes live only if it wins at every horizon. Recalibrating the drift thresholds cut false alarms from 100% to 6% of weeks.

ResearchForge AI: Multi-Agent Research Assistant Live Demo | GitHub
Python, FastAPI, LangGraph, PostgreSQL, Redis, Next.js, Docker

- Built a 7-agent LangGraph pipeline that plans searches, pulls papers from arXiv, Semantic Scholar and Crossref, filters them for relevance and writes cited summaries, checked by fact-checking, debate and self-critique steps. Cut runtime from 3–4 minutes to about 1 minute by reducing LLM calls from 15 to 7 and running independent agents in parallel.
- Added JWT/bcrypt login, per-user data isolation, Redis rate limiting and live progress updates, deployed at $0/month on Vercel, Render and Neon. GitHub Actions runs linting, type checks, dependency scans and 266 tests, including Postgres integration tests.

Achievements

- **Best Pitch Award**, Kalyana Karnataka Startup Conclave 2024: chosen over 100+ teams for an AI-based pregnancy-monitoring belt.

Technical Skills

Languages: Python, SQL
ML / DL & Data: Scikit-learn, LightGBM, PyTorch, TensorFlow, Keras, Pandas, NumPy, Matplotlib
GenAI & LLMs: LLM Evaluation, Model Benchmarking, Prompt Engineering, RAG, LangGraph, LangChain, Multi-Agent Systems
Backend & MLOps: FastAPI, PostgreSQL, Redis, Docker, GitHub Actions, Evidently, MLflow, REST APIs
Tools: Git, Jupyter, MySQL, Vercel, Render

Education

B.E., Artificial Intelligence and Machine Learning Aug 2022 – Jun 2026
P.D.A. College of Engineering (VTU), Kalaburagi | CGPA: 7.89/10

Certifications

Oracle Cloud Infrastructure Data Science Professional (Oracle) · AI Fundamentals (Google) · Python with Data Science (Ethnotech) · Internet of Things (NPTEL) · Advanced C Programming (Cisco)